

# Complex Plane Representation of Weak-Isospin Projections and Flavour–Chirality Partition

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## Abstract

We develop a geometric representation of the Left–Right (LR) symmetric electroweak structure by treating the third components of left- and right-handed weak isospin,  $T_3^L$  and  $T_3^R$ , as orthogonal coordinates of an internal two-dimensional space. The complex coordinate

$$T_3 = T_3^L + iT_3^R$$

provides a compact representation of this weak-isospin plane, while the invariant norm

$$|T_3|^2 = (T_3^L)^2 + (T_3^R)^2$$

defines the ideal internal manifold for fundamental fermion representations in the minimal Left–Right symmetric framework. An invertible transformation to symmetric and antisymmetric coordinates,

$$P = T_3^L + T_3^R, \quad D = T_3^L - T_3^R,$$

reveals two complementary geometric sectors of the same internal structure. The symmetric coordinate  $P$  reproduces the electroweak charge contribution through the generalized Gell–Mann–Nishijima relation,

$$Q = P + \frac{B - L}{2},$$

and provides a chirality-independent classification of up-type and down-type quark sectors. The antisymmetric coordinate leads to a geometric chirality index,

$$\chi_g = \frac{P}{D},$$

which distinguishes the left- and right-handed orientations of the fundamental Left–Right states. The formulation demonstrates that flavour-sector classification and left–right orientation can be viewed as complementary projections of a single weak-isospin geometry, without introducing additional gauge quantum numbers. We further explore departures from the ideal internal configuration through a complex deformation,

$$\Delta T_3 = \delta e^{i\phi},$$

which provides geometric parameters describing the magnitude and orientation of internal weak-isospin displacements. Such deformations offer a possible framework for investigating flavour-dependent effects and complex phases associated with quark mixing. The resulting construction provides a geometric interpretation of the Left–Right weak-isospin structure and establishes a basis for exploring possible connections between internal electroweak geometry, flavour mixing, and CP-sensitive phenomena.

# 1 Introduction

The Standard Model (SM) successfully describes electroweak interactions through the gauge symmetry  $SU(2)_L \otimes U(1)_Y$  [1–3], where the third component of weak isospin,  $T_3^L$ , together with the hypercharge  $Y$ , determines the electric charge through the Gell-Mann–Nishijima relation [4, 5],

$$Q = T_3^L + \frac{Y}{2}. \quad (1)$$

Within the SM, the electroweak interaction possesses an intrinsic chiral structure. Left-handed fermions transform as doublets under  $SU(2)_L$  and therefore carry non-zero weak-isospin projections  $T_3^L = \pm 1/2$ , whereas right-handed fermions are singlets under  $SU(2)_L$ . Consequently, charged weak interactions couple directly only to left-handed fermion fields.

Left–Right (LR) symmetric theories extend this structure by introducing a corresponding right-handed weak interaction sector [6–8],

$$\mathcal{G}_{LR} = SU(3)_C \otimes SU(2)_L \otimes SU(2)_R \otimes U(1)_{B-L}, \quad (2)$$

where left- and right-handed fermions appear as complementary representations. In the LR symmetric framework, the hypercharge relation becomes [7, 8]

$$\frac{Y}{2} = T_3^R + \frac{B-L}{2}, \quad (3)$$

and the electric charge takes the generalized LR Gell-Mann–Nishijima form

$$Q = T_3^L + T_3^R + \frac{B-L}{2}. \quad (4)$$

Equation (4) shows that the electroweak contribution to electric charge is determined by the pair of commuting generators  $T_3^L$  and  $T_3^R$ . This observation motivates the interpretation of these two weak-isospin projections as coordinates of an internal two-dimensional space.

The purpose of this work is to investigate the geometric structure associated with these coordinates. We show that the pair  $(T_3^L, T_3^R)$  admits a natural decomposition into symmetric and antisymmetric sectors. The symmetric combination provides a flavour-related classification through the generalized Gell-Mann–Nishijima relation, while the antisymmetric sector provides a geometric characterization of left–right orientation. The resulting formulation does not introduce new gauge generators or alter the LR algebra. Instead, it provides an internal coordinate representation in which flavour classification and chiral orientation appear as complementary projections of the same weak-isospin structure. The paper develops this construction through a sequence of theorems: internal weak-isospin geometry, symmetric–antisymmetric decomposition, flavour classification, geometric chirality classification, and their complementarity.

## 2 Weak-Isospin Decomposition

### 2.1 Quantum Chiral Mixing and Charge-Preserving Internal Evolution

For a massive fermion, the Dirac mass term couples the left- and right-chiral components [9],

$$m\bar{\psi}\psi = m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L), \quad (5)$$

leading to the equations of motion

$$i \not{\partial} \psi_L = m\psi_R, \quad i \not{\partial} \psi_R = m\psi_L. \quad (6)$$

Therefore, although chirality remains a well-defined field-theoretic projection, the quantum state of a massive fermion may contain time-dependent left- and right-chiral amplitudes. A physical state may be represented schematically as

$$|\Psi_f(t)\rangle = \cos \alpha_f(t) \left| \frac{s_f}{2}, 0 \right\rangle + e^{i\phi_f(t)} \sin \alpha_f(t) \left| 0, \frac{s_f}{2} \right\rangle, \quad (7)$$

where  $s_f = \pm 1$  specifies the flavour orientation and  $\alpha_f(t)$  describes the relative chiral composition. The generalized Gell-Mann–Nishijima relation is preserved as an expectation-value relation,

$$\langle Q \rangle = \langle \hat{T}_3^L \rangle(t) + \langle \hat{T}_3^R \rangle(t) + \frac{B-L}{2}. \quad (8)$$

For the evolving state,

$$\langle \hat{T}_3^L \rangle = \frac{s_f}{2} \cos^2 \alpha_f(t), \quad (9)$$

and

$$\langle \hat{T}_3^R \rangle = \frac{s_f}{2} \sin^2 \alpha_f(t). \quad (10)$$

Hence,

$$\langle Q \rangle = \frac{s_f}{2} (\cos^2 \alpha_f + \sin^2 \alpha_f) + \frac{B-L}{2}, \quad (11)$$

which gives

$$\langle Q \rangle = \frac{s_f}{2} + \frac{B-L}{2}. \quad (12)$$

Thus the electric charge remains invariant even though the internal left–right composition of the quantum state may evolve. The complex weak-isospin coordinate is therefore interpreted as

$$T_3 = \langle \hat{T}_3^L \rangle + i \langle \hat{T}_3^R \rangle, \quad (13)$$

representing the observable internal projection of the fermionic state. The generators themselves remain fixed operators of the  $SU(2)_L$  and  $SU(2)_R$  algebras.

## 2.2 Weak-Isospin Representation in a Complex Plane

The generators  $T_3^L$  and  $T_3^R$  belong to commuting algebras,

$$[T_i^L, T_j^R] = 0,$$

and may therefore be represented as orthogonal coordinates of an internal two-dimensional space.

**Theorem 1** (Internal Weak-Isospin Geometry). *For fundamental fermion doublet representations of the minimal Left–Right symmetric theory, the internal coordinates satisfy*

$$T_3 = T_3^L + iT_3^R, \quad (14)$$

with invariant norm

$$|T_3|^2 = (T_3^L)^2 + (T_3^R)^2. \quad (15)$$

For the undeformed fundamental states,

$$(T_3^L)^2 + (T_3^R)^2 = \frac{1}{4}. \quad (16)$$

Thus the allowed ideal configurations lie on an invariant circle of radius  $1/2$  in the internal weak-isospin plane.

*Proof.* In the minimal LR representation, each fundamental fermion carries a non-zero weak-isospin projection in either the left or right sector,

$$\left(\pm\frac{1}{2}, 0\right), \quad \left(0, \pm\frac{1}{2}\right).$$

All such configurations satisfy

$$(T_3^L)^2 + (T_3^R)^2 = \frac{1}{4}.$$

Since the two generators commute, they define a two-dimensional internal coordinate space, and the complex representation provides a convenient parametrization of this space.  $\square$

### 3 Chiral and Flavour Partitioning

#### 3.1 Symmetric and Antisymmetric Decomposition

The two-dimensional weak-isospin coordinate space admits an alternative basis obtained from symmetric and antisymmetric combinations of the two commuting projections. We define

$$\begin{pmatrix} P \\ D \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} T_3^L \\ T_3^R \end{pmatrix}, \quad (17)$$

or explicitly,

$$P = T_3^L + T_3^R, \quad (18)$$

and

$$D = T_3^L - T_3^R. \quad (19)$$

**Theorem 2** (Symmetric–Antisymmetric Decomposition). *The internal weak-isospin coordinate system admits an invertible change of basis from  $(T_3^L, T_3^R)$  to the complementary coordinates  $(P, D)$ . The inverse transformation is*

$$T_3^L = \frac{P + D}{2}, \quad T_3^R = \frac{P - D}{2}. \quad (20)$$

*The transformation reorganizes the same physical information into symmetric and antisymmetric sectors and does not introduce additional quantum numbers.*

*Proof.* The determinant of the transformation matrix is

$$\det \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = -2 \neq 0,$$

therefore the transformation is invertible. Direct substitution gives Eq. (20).  $\square$

Using Eq. (4), the electric charge relation becomes

$$Q = P + \frac{B - L}{2}. \quad (21)$$

Thus  $P$  represents the complete weak-isospin contribution to the electric charge.

### 3.2 The Flavour Sector

The symmetric coordinate  $P$  provides a natural classification of quark states according to their electroweak charge sector.

**Theorem 3** (Electroweak Flavour Classification). *For quarks, the symmetric coordinate*

$$P = T_3^L + T_3^R$$

*is independent of chirality and satisfies*

$$P = \begin{cases} +\frac{1}{2}, & \text{up-type quarks } (u, c, t)_{L,R}, \\ -\frac{1}{2}, & \text{down-type quarks } (d, s, b)_{L,R}. \end{cases} \quad (22)$$

*Proof.* All quarks possess  $B - L = 1/3$ . For an up-type quark,  $Q = +2/3$ , and therefore from Eq. (21),

$$P = Q - \frac{B - L}{2} = \frac{2}{3} - \frac{1}{6} = \frac{1}{2}.$$

For a down-type quark,  $Q = -1/3$ , giving

$$P = -\frac{1}{3} - \frac{1}{6} = -\frac{1}{2}.$$

Because electric charge and  $(B - L)$  are independent of chirality, the value of  $P$  is identical for left- and right-handed members of the same electroweak charge sector.  $\square$

*Remark 4.* The coordinate  $P$  is chirality-independent. It provides a geometric separation between the up-type and down-type electroweak charge sectors.

### 3.3 The Geometric Chirality Sector

The antisymmetric coordinate  $D$  contains the relative orientation between left and right weak-isospin projections. This motivates the definition of a geometric chirality index.

**Theorem 5** (Geometric Chirality Classification). *Define*

$$\chi_g = \frac{P}{D} = \frac{T_3^L + T_3^R}{T_3^L - T_3^R}. \quad (23)$$

*For the ideal LR fermion states,*

$$\chi_g = \begin{cases} +1, & \text{left-handed states,} \\ -1, & \text{right-handed states.} \end{cases} \quad (24)$$

*Proof.* A left-handed state has  $T_3^R = 0$ ,  $T_3^L = \pm 1/2$ . Hence  $P = T_3^L$ ,  $D = T_3^L$ , and  $\chi_g = 1$ .

A right-handed state satisfies  $T_3^L = 0$ ,  $T_3^R = \pm 1/2$ , giving  $P = T_3^R$ ,  $D = -T_3^R$ , and  $\chi_g = -1$ .  $\square$

*Remark 6* (Non-singularity of the geometric index). The only possible singularity occurs when  $D = T_3^L - T_3^R = 0$ . For the physical quark states considered here, the LR representation contains a non-zero projection in only one sector, so  $D \neq 0$ . Hence the geometric index is well defined throughout the fundamental quark state space.

*Remark 7.* The geometric index  $\chi_g$  does not replace the chirality operator of quantum field theory. Rather, it provides an internal coordinate representation of the left-right orientation encoded by the LR weak-isospin projections.

### 3.4 Complementarity of Flavour and Chirality

**Theorem 8** (Complementarity). *The electroweak charge classification and left–right orientation of a fundamental quark state are complementary projections of the same internal weak-isospin coordinates. The quantities  $P$  and  $\chi_g$  are simultaneously obtained from the pair  $(T_3^L, T_3^R)$  and together reconstruct the original internal coordinates.*

*Proof.* The coordinate  $P$  depends on the symmetric combination  $T_3^L + T_3^R$ , whereas  $\chi_g$  depends on the ratio  $(T_3^L + T_3^R)/(T_3^L - T_3^R)$ . Thus  $P$  determines the charge sector while  $\chi_g$  determines the left–right orientation. From  $D = P/\chi_g$ , the original coordinates are recovered as

$$T_3^L = \frac{P}{2} \left( 1 + \frac{1}{\chi_g} \right), \quad (25)$$

and

$$T_3^R = \frac{P}{2} \left( 1 - \frac{1}{\chi_g} \right). \quad (26)$$

Therefore the pair  $(P, \chi_g)$  contains the same information as the original weak-isospin coordinates.  $\square$

## 4 Weak-Isospin Activity, Internal Deformation, and CP-Sensitive Geometry

The preceding sections established that the pair of commuting weak-isospin coordinates  $(T_3^L, T_3^R)$  provides an internal representation of the Left–Right electroweak structure. The symmetric coordinate  $P = T_3^L + T_3^R$  determines the electroweak charge sector, while the geometric chirality index  $\chi_g = P/D$  characterizes the left–right orientation of the state.

The ideal LR-symmetric configurations satisfy  $|T_3| = 1/2$ , corresponding to the invariant weak-isospin manifold introduced in Theorem 1. We now consider possible departures from this idealized configuration.

**Postulate 1** (Weak-Isospin Activity Principle). *The internal electroweak configuration of a fermion may be characterized by an effective weak-isospin activity associated with the available left- and right-handed components. In the ideal symmetric limit, this activity places the internal coordinate on the invariant manifold  $|T_3| = 1/2$ .*

A departure from the ideal configuration is represented by an internal deformation,

$$T_3 \rightarrow T_3 + \Delta T_3. \quad (27)$$

Such a deformation changes the location of the state within the internal weak-isospin plane. The magnitude of the deformation is not, by itself, a CP-violating quantity; rather, the phase structure associated with a complex deformation may provide a geometric representation of CP-sensitive effects.

For massive fermions the weak-isospin coordinates are interpreted as expectation values,

$$T_3 = \langle \hat{T}_3^L \rangle + i \langle \hat{T}_3^R \rangle. \quad (28)$$

The deformation is therefore represented by the complex vector

$$\Delta T_3 = \delta e^{i\phi}, \quad (29)$$

where  $\delta$  measures the magnitude and  $\phi$  specifies the orientation of the deformation in the internal plane. The corresponding Cartesian components are

$$\Delta T_3^L = \delta \cos \phi, \quad (30)$$

and

$$\Delta T_3^R = \delta \sin \phi. \quad (31)$$

Hence the deformed internal coordinate becomes

$$T_3' = (T_3^L + \delta \cos \phi) + i(T_3^R + \delta \sin \phi). \quad (32)$$

The magnitude of the internal activity deformation may be described by an effective weak-isospin activity factor,

$$\Gamma = 1 + (T_3^L)' + (T_3^R)'. \quad (33)$$

The normalization is chosen such that the undeformed reference configuration has a convenient unit baseline. Substituting the deformed coordinates gives

$$\Gamma = 1 + T_3^L + T_3^R + \delta(\cos \phi + \sin \phi). \quad (34)$$

For deformations dominated by a single principal direction, the projected effect may be represented by an effective parameter  $\delta_b$ ,

$$\Gamma = 1 + T_3^L + T_3^R + \delta_b. \quad (35)$$

The factor  $\Gamma$  therefore characterizes the effective change of the internal weak-isospin configuration.

A complementary normalization factor may be defined from the hypercharge sector,

$$K = 1 + \frac{Y}{2}. \quad (36)$$

Using the Left–Right hypercharge relation (3), this becomes

$$K = 1 + T_3^R + \frac{B - L}{2}. \quad (37)$$

Whereas  $\Gamma$  describes the response of the internal weak-isospin configuration,  $K$  represents a corresponding background normalization associated with the electroweak quantum numbers.

The departure from the ideal weak-isospin manifold is measured by the geometric deformation parameter,

$$\Delta_{\text{geom}} = |T_3| - \frac{1}{2}. \quad (38)$$

A non-zero  $\Delta_{\text{geom}}$  indicates a departure from the ideal circle. If the deformation possesses a non-trivial complex phase, the orientation of this deformation may provide a geometric representation of CP-sensitive structure.

The quantities  $(\delta, \phi, \Gamma, K)$  therefore provide a geometric description of internal weak-isospin deformation. Their possible connection with flavour mixing and CP violation is explored through the CKM correspondence developed in the following section.

## 5 Geometric Correspondence with CKM Mixing Parameters

The Cabibbo–Kobayashi–Maskawa (CKM) matrix provides the standard description of quark flavour mixing and CP violation within the Standard Model [10, 11]. In the Wolfenstein parametrization [12], the complex structure of the CKM matrix is expressed through the parameters  $(\lambda, A, \bar{\rho}, \bar{\eta})$ , where  $\bar{\rho}$  and  $\bar{\eta}$  determine the position of the apex of the unitarity triangle. These parameters are obtained from global fits to experimental measurements [13] and are not derived from an underlying geometric construction within the Standard Model.

The internal weak-isospin geometry developed in this work provides a possible geometric interpretation of the origin of complex flavour parameters. The internal deformation introduced in the previous section is represented by

$$\Delta T_3 = \delta e^{i\phi}, \quad (39)$$

with magnitude  $\delta$  and orientation  $\phi$  in the internal  $(T_3^L, T_3^R)$  plane. The normalized geometric components are defined as

$$\rho_g = \frac{\Delta T_3^L}{\delta}, \quad (40)$$

and

$$\eta_g = \frac{\Delta T_3^R}{\delta}. \quad (41)$$

By construction,

$$\rho_g^2 + \eta_g^2 = 1, \quad (42)$$

which expresses the fact that  $(\rho_g, \eta_g)$  specify only the orientation of the deformation vector. The internal deformation can therefore be written as

$$\Delta T_3 = \delta(\rho_g + i\eta_g), \quad (43)$$

and the deformed weak-isospin coordinate becomes

$$T_3' = T_3 + \delta(\rho_g + i\eta_g). \quad (44)$$

The angle associated with the deformation is

$$\phi = \tan^{-1}\left(\frac{\eta_g}{\rho_g}\right), \quad (45)$$

whereas the magnitude  $\delta$  controls the radial departure from the ideal weak-isospin configuration.

The geometric quantities  $(\rho_g, \eta_g)$  should be distinguished from the phenomenological Wolfenstein parameters  $(\bar{\rho}, \bar{\eta})$  [12]. The former describe the normalized orientation of an internal weak-isospin deformation, whereas the latter are experimentally extracted parameters describing the CKM mixing structure [13]. A possible correspondence may therefore be written schematically as

$$(\rho_g, \eta_g, \delta) \longrightarrow (\bar{\rho}, \bar{\eta}, \text{CKM deformation scale}), \quad (46)$$

where the precise mapping requires the dynamical relation between the internal weak-isospin deformation and the quark mixing matrix.

In this interpretation, the complex phase entering flavour mixing is associated with the orientation of the internal deformation vector. The weak-isospin activity factor  $\Gamma$  determines the effective strength of the deformation, while the normalized coordinates  $(\rho_g, \eta_g)$  determine its direction. Thus the CKM complex structure may be viewed as an observable projection of an underlying internal weak-isospin geometry. The present formulation does not replace the phenomenological CKM parametrization, but provides a geometric framework in which the appearance of mixing angles and complex phases may be investigated. A complete derivation of the CKM matrix elements from the internal deformation parameters requires an additional dynamical model and is left for future development.

## 6 Conclusions

In this work, we have investigated the internal geometric structure of the LR symmetric Gell-Mann–Nishijima relation. Starting from the standard LR electroweak framework [6–8],

$$Q = T_3^L + T_3^R + \frac{B - L}{2},$$

we have shown that the two commuting weak-isospin projections  $(T_3^L, T_3^R)$  may be represented as coordinates of an internal two-dimensional space. The complex representation

$$T_3 = T_3^L + iT_3^R$$

provides a compact description of this internal space. For the fundamental fermion representations of the minimal Left–Right symmetric theory, the undeformed configurations satisfy the invariant condition

$$(T_3^L)^2 + (T_3^R)^2 = \frac{1}{4},$$

corresponding to an internal weak-isospin manifold of radius  $1/2$ .

An invertible transformation to the symmetric and antisymmetric coordinates  $P = T_3^L + T_3^R$  and  $D = T_3^L - T_3^R$  provides a complementary representation of the same internal information. The symmetric coordinate  $P$  determines the electroweak charge sector through  $Q = P + (B - L)/2$  and separates the up-type and down-type quark sectors. The antisymmetric coordinate provides the geometric chirality index  $\chi_g = P/D$ , which reproduces the left–right orientation of the fundamental LR fermion states.

The resulting picture is that charge-sector classification and left–right orientation are not independent geometric constructions, but complementary projections of the same weak-isospin coordinate system. The formulation therefore provides an internal geometric organization of flavour classification and chirality orientation within the LR framework.

We further investigated departures from the ideal invariant manifold through the internal deformation  $T_3 \rightarrow T_3 + \Delta T_3$ . Such a deformation introduces a magnitude and orientation,  $\Delta T_3 = \delta e^{i\phi}$ , which provide geometric parameters for describing departures from the ideal LR configuration. The associated activity factor and normalized deformation coordinates offer a possible framework for studying how internal weak-isospin structure may contribute to flavour mixing and complex phases.

The connection with CKM phenomenology [10–12] is therefore interpreted as a geometric correspondence rather than a replacement of the standard CKM parametrization. The normalized deformation coordinates describe the orientation of an internal weak-isospin displacement, while the phenomenological Wolfenstein parameters remain experimentally determined quantities [13] whose relation to the internal geometry requires an additional dynamical construction.

The radial deformation introduced for the right-handed bottom quark illustrates how a localized modification of the internal coordinate can preserve both the electroweak charge classification and the geometric chirality index. This suggests that internal weak-isospin deformations may provide a useful direction for exploring the origin of flavour-dependent effects.

The main result of this work is therefore a geometric organization of the LR weak-isospin structure, such as, flavour classification, chirality orientation, and possible internal deformation are viewed as different aspects of a common two-dimensional weak-isospin geometry. Further work is required to determine whether the proposed deformation variables can be connected dynamically to the observed pattern of CKM mixing and CP violation.

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